

Template Week 3 – Hardware

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Assignment 3.1: Examine your phone

What processor is in your phone?

- Apple A18 Pro

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

- The Apple A18 Pro belongs to the ARM architecture family. It uses the ARM64 (AArch64) instruction set, which is an Armv9-class architecture. This architecture is designed for high performance with lower consumption which is important for mobile devices.

How much RAM is in it?

- 8GB. With this amount of RAM allows multiple applications to run smoothly at the same time without being closed by the operating system.

How much storage does your phone have?

- 256GB. This amount of storage provides enough space for applications, photos, videos, and system files.

What operating system is running on your phone?

- iOS 18

Approximately how many applications do you have installed?

- 114 applications

Which application do you use the most?

- Spotify

Can your phone be charged with what type of plug?

- USB C. The USB-C supports both charging and data transfer and is the current standard for modern smartphones.

Which I/O ports can you visually see on your phone?

- USB-C port (charging and data), speaker grilles (audio output), microphone openings (audio input), SIM tray (mobile network connectivity).

Assignment 3.2: Examine your laptop

What processor is in your laptop?

- Intel Core i7-8750H

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

- The Intel Core i7-8750H belongs to the x86 architecture family and uses the x86-64 (64-bit) instruction set. This architecture is commonly used in laptops and desktop computers and is optimized for high performance.
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How much RAM is in it?

- 16 GB.
- This amount of RAM allows multiple applications to run simultaneously without significant performance loss.

How much storage does your laptop have?

- 256 GB SSD
- 1 TB HDD
- The SSD is mainly used for the operating system and applications, while HDD is used for larger data storage.

Which operating system is running on your laptop?

- Windows 11 Education

Approximately how many applications do you have installed?

- 84 Applications

Which application do you use the most?

- Google Chrome

Can your laptop be charged with what type of plug?

- DC barrel-type connector
- This connector supplies power directly to the laptop and is commonly used on non-USB-C charging devices.

Which I/O ports can you visually see on your laptop?

- DC charging port, Ethernet (RJ-45) port, USB Type-A ports, USB-C port, HDMI port, Microphone/headphone combo jack, SD card reader, Kensington lock slot.
- Some of these interfaces support both input and output depending on their usage (for example USB and audio connections).

Assignment 3.3: Power to the laptop

What is the input voltage?

- 100–240V ~ 50–60Hz 2.0A
- This means the power adapter can be used with different wall voltages worldwide.

What is the output voltage?

- $19V = 6.32A$

How many watts can your power adapter deliver?

- $19V \times 6.32A = \sim 120W$
- This is the maximum power the adapter can supply to the laptop.

Is the input voltage AC or DC?

- AC (Alternating Current)

Is the output voltage AC or DC?

- DC (Direct Current)

AC/DC what is that?

(AC) Alternating Current.

- AC is electricity that continuously changes direction.
- This is the type of power supplied by wall outlets.
- In Europe, the direction changes 50 times per second (50Hz)

(DC) Direct Current

- DC is electricity that flows in one direction only.
- Electronic devices require stable power, which is why laptops, phones, and batteries use DC.
- The power adapter converts AC from the wall outlet into DC for the laptop.

If you reverse the polarity of the output voltage, is that bad for your laptop?

- Yes. Reversing the polarity is harmful because the internal electronics are designed for a specific current direction.
- Incorrect polarity can cause permanent damage to components

You forgot your power adapter; your laptop normally needs 15 watts. You will be loaned a power adapter that can deliver 50 watts. Voltage, polarity, etc. are all the same compared to the original power adapter. You can connect the borrowed power adapter to your laptop. What will happen? Also explain why you think that.

- Yes, the laptop can safely be connected.
- The laptop will only draw the amount of power it needs (15 W), even though the adapter can supply up to 50 W.
- As long as the voltage and polarity match, a higher wattage adapter does not damage the laptop.
- The extra available power simply remains unused.

Assignment 3.4: Build your dream PC

Screenshots PC configuration + motivation:

CPU



AMD Ryzen 7 7700 3.6 GHz 8-Core Processor

Base	Promo	Shipping	Tax	Availability		Price	Where	
€279.00	—	FREE	—	In stock	⚙️	€279.00	AZERTY	Buy ✕

CPU Cooler



Noctua NH-D15 chromax.black 82.52 CFM CPU Cooler

Base	Promo	Shipping	Tax	Availability		Price	Where	
€118.49	—	 Prime	—	In stock	⚙️	€118.49	amazon.nl	Buy ✕

Motherboard



Gigabyte B650 EAGLE AX ATX AM5 Motherboard

Base	Promo	Shipping	Tax	Availability		Price	Where	
€129.90	—	 Prime	—	In stock	⚙️	€129.90	amazon.nl	Buy ✕

Memory



G.Skill Trident Z5 Neo 32 GB (2 x 16 GB) DDR5-6000 CL32 Memory

Base	Promo	Shipping	Tax	Availability		Price	Where	
€189.99	—	 Prime	—	In stock	⚙️	€189.99	amazon.nl	Buy ✕

Storage



Kingston NV3 500 GB M.2-2280 PCIe 4.0 X4 NVME Solid State Drive

Base €81.90	Promo —	Shipping €5.95	Tax —	Availability In stock	⚙️	Price €87.85	Where ALTERNATE	Buy	×
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Storage



Kingston NV3 1 TB M.2-2280 PCIe 4.0 X4 NVME Solid State Drive

Base €116.90	Promo —	Shipping FREE	Tax —	Availability In stock	⚙️	Price €116.90	Where ALTERNATE	Buy	×
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Video Card



MSI GeForce RTX 3060 Ventus 2X 12G GeForce RTX 3060 12GB 12 GB Video Card

Base €565.50	Promo —	Shipping —	Tax —	Availability In stock	⚙️	Price €565.50	Where amazon.nl	Buy	×
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Case



Corsair 3500X ARGB ATX Mid Tower Case

Base €128.97	Promo —	Shipping Prime	Tax —	Availability In stock	⚙️	Price €128.97	Where amazon.nl	Buy	×
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Power Supply



MSI MAG A750GL PCIE5 750 W 80+ Gold Certified Fully Modular ATX Power Supply

Base €119.03	Promo —	Shipping —	Tax —	Availability In stock	⚙️	Price €119.03	Where amazon.nl	Buy	×
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Operating System



Microsoft Windows 11 Home OEM - DVD 64-bit

Base	Promo	Shipping	Tax	Availability		Price	Where	
€126.00	—	FREE	—		⚙️	€126.00	PARADIGIT	Buy ×

Monitor



KOORUI 24E3 24.0" 1920 x 1080 165 Hz Monitor

Base	Promo	Shipping	Tax	Availability		Price	Where	
—	—	—	—	—	⚙️	No Prices Available	—	Buy ×

This PC is designed as a general-purpose system that is suitable for everyday use such as schoolwork, browsing, and media use, while also being powerful enough to support future projects that may require more performance.

The selected components provide a good balance between performance and practicality. The processor is suitable for multitasking and development-related work, while the graphics card can handle graphical tasks without being excessive.

The system includes a **32 GB** of RAM, which helps ensure smooth operation when multiple applications are running at the same time. The storage setup combines a fast SSD for the operating system and programs with additional storage for files and projects.

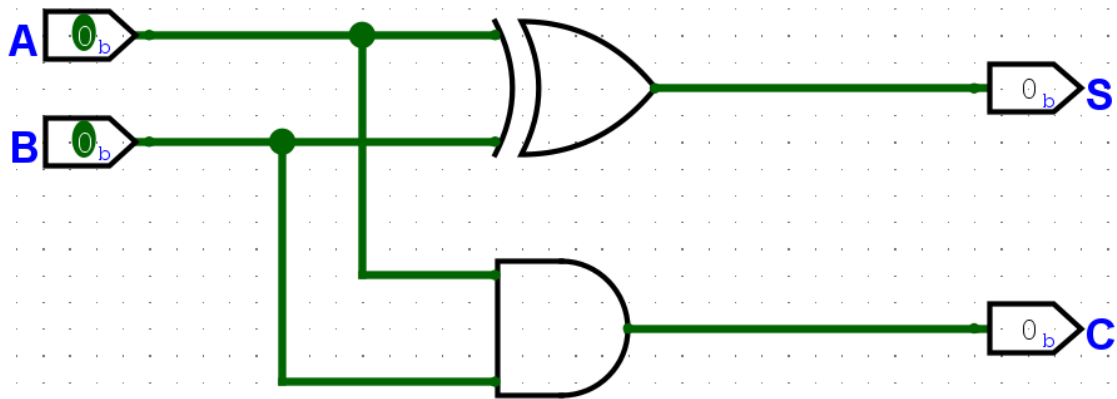
Overall, this build focuses on reliability rather than maximum performance, making it suitable for long-term use.

Assignment 3.5: Adders

Complete the **half adder**, **full adder** and **4-bit adder** assignment as described in the PowerPoint slides of week 3 in Logisim. Save the chip design and export three PNG pictures of the separate finished designs. See the PowerPoint slides of week 3.

Paste the three exported PNG pictures in here.

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